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Evaluating Parameter-Efficient Fine-Tuning vs. Prompt Engineering for Profile-Driven Recommendations

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Abstract

Large language models (LLMs) show significant promise for personalized recommendation systems, yet identifying the most effective adaptation strategy for structured data remains a challenge. This study evaluates the performance trade-offs between parameter-efficient fine-tuning (PEFT) and in-context learning approaches for personalized item recommendation. We utilized explicit user profiles, encompassing specific likes and dislikes, alongside a comprehensive JSON-formatted item catalog. Two distinct adaptation paradigms were compared: PEFT, specifically utilizing Low-Rank Adaptation (LoRA) and Infused Adapter by Inhibiting and Amplifying Inner Activations (IA^3), against robust prompt engineering strategies optimizing input instructions without altering model weights. The approaches were systematically evaluated using standard ranking metrics, including Normalized Discounted Cumulative Gain (NDCG@K) and Hit Ratio (HR@K), alongside computational efficiency measures. Our findings demonstrate that while prompt engineering establishes a competitive baseline, PEFT methods, particularly LoRA, achieve superior alignment with nuanced user preferences, yielding significant improvements in ranking accuracy. Furthermore, PEFT effectively mitigates context-window limitations, reducing inference latency compared to extensive prompt demonstrations. Consequently, we conclude that adapter-based fine-tuning provides more reliable and scalable personalized recommendations from structured catalogs than prompt engineering alone, successfully bridging raw LLM reasoning with strict domain-specific requirements.

Keywords: large language models; parameter-efficient fine-tuning; prompt engineering; recommender systems; Low-Rank Adaptation (LoRA); structured data processing; in-context learning

1. Introduction

The integration of Large Language Models (LLMs) into recommendation systems has fundamentally transformed the field, treating recommendation as a sequence generation task. However, adapting these models to dynamic user preferences presents significant challenges regarding stability and plasticity. To address the continuous evolution of users and items, recent research has explored parameter-efficient fine-tuning (PEFT) methods like Low-Rank Adaptation (LoRA); for instance, the PESO framework maintains a single evolving adapter anchored by a proximal term to dynamically capture new user behaviors while mitigating the catastrophic forgetting associated with sequential updates [1].

Transitioning LLMs into conversational recommenders introduces additional complexities, particularly concerning semantic misalignment and cumbersome training processes.

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To streamline this, the PECRS framework reformulates the problem entirely as a natural language processing task, utilizing a frozen pre-trained language model and rich textual metadata to perform both item recommendation and response generation. This unified architecture can be optimized in a single training stage using PEFT techniques, significantly reducing computational overhead [2].

While efficiency is crucial, standard PEFT methods often lack parameter-level personalization because they rely on a static, shared LoRA module for all users. The RecLoRA framework addresses this by introducing a Personalized LoRA module, which dynamically generates unique weights for each user through a soft routing mechanism guided by a conventional recommendation model [3].

Beyond parameter adaptation, prompt engineering and in-context learning offer alternative pathways for efficient LLM integration. Because routing all users through an LLM is computationally prohibitive, recent hybrid task allocation frameworks propose identifying "weak users" with sparse interaction histories and contextualizing their data into natural language instructions. This targeted approach drastically reduces token usage and latency while improving recommendation quality where traditional systems struggle [4].

Finally, applying these adaptations in complex, multi-domain industrial settings requires balancing general user behavior with domain-specific data sparsity. Frameworks such as MLoRA achieve this by combining a shared backbone, which learns generalizable information across domains, with specialized low-rank adaptive modules tailored to capture distinctive, domain-specific features [5].

2. Background and Related Work

The integration of Large Language Models (LLMs) into recommender systems has been systematically categorized by Elmoghazy et al., who established a taxonomy based on the LLM's specific role, learning paradigms, and overarching system architecture. Their comparative study emphasizes that while LLMs can act as textual feature encoders or direct recommendation generators, they face persistent challenges such as context window constraints, computational latency, and a disparity between language objectives and recommendation tasks [6]. Further structuring this domain, Bao et al. surveyed the field by dividing architectures into Discriminative and Generative paradigms. Their research details how generative models provide impressive zero-shot capabilities and natural language explainability, despite struggling with context length limits and inherent biases [7]. Additionally, Liu et al. explored language modeling paradigm adaptations, contrasting traditional pre-training with parameter-efficient prompt learning to overcome data sparsity, while noting ongoing issues with factual consistency and user privacy [8].

Expanding on these foundational taxonomies, Nawara and Kashef provided a structured examination of how LLM-powered systems overcome conventional recommendation limitations like data sparsity, cold-start issues, and long-tail challenges. Their survey categorizes recommenders into five distinct paradigms—discriminative, generative, hybrid, graph-enhanced, and multi-modal—while evaluating the transition from traditional metrics to modern generative assessments and "LLM-as-a-Judge" frameworks [9]. Similarly, Shehmir and Kashef conducted a comprehensive systematic literature review of over 150 works to analyze the integration of transformer-based backbones such as BERT, GPT, and T5 within recommendation contexts. Their findings map the application of prompt engineering and parameter-efficient fine-tuning (PEFT) across diverse applications while highlighting persistent algorithmic hurdles like high inference latency and societal bias [10].

Shifting the focus specifically toward personalized environments, Chen et al. detailed how the integration of LLMs transforms user experiences from passive filtering into active, conversational, and explainable processes. They established a multifaceted taxonomy where LLMs act as content interpreters, system reasoners, conversational agents, and tool-learning controllers, though they noted significant operational constraints regarding industrial deployment costs and the "phantom" problem of hallucinated content [11]. To frame the technical necessity of tailoring these models, Ling et al. explored the mechanisms of domain specialization required to transform general-purpose LLMs into disruptive tools for highly specific fields. Their systematic framework categorizes adaptation techniques based on model accessibility into black-box external augmentation, grey-box prompt crafting, and white-box model fine-tuning, emphasizing that while fine-tuning offers deep alignment with domain objectives, it introduces risks of catastrophic forgetting and high resource overhead [12].

Transitioning from theoretical surveys to empirical frameworks, Xu et al. proposed LLMRec to systematically evaluate the performance of LLMs functioning directly as recommender systems. Their extensive empirical analysis demonstrated that fine-tuning open-source models via parameter-efficient methods like Low-Rank Adaptation (LoRA) yields superior domain adaptation compared to zero-shot approaches, although targeted reasoning prompts like Chain-of-Thought remain crucial for optimizing prediction accuracy [13].

Focusing specifically on the advancement of parameter-efficient methods, Zhu et al. addressed the inefficiency of static LoRA matrices by introducing RecLoRA. This framework dynamically aggregates multiple independent meta-LoRA weights to create a unique, personalized module for each user, successfully balancing recommendation effectiveness and processing time [3]. Building further on personalized adaptation, Kim et al. identified that standard PEFT foundation models often generate user-blind item embeddings. To resolve this, they proposed PerPEFT, a framework that groups users by similar interests and assigns a distinct, dedicated PEFT module to each demographic. By employing group-specific hard negative sampling, their architecture dynamically captures fine-grained item characteristics, proving highly adaptable to various structures like LoRA and IA^3 with minimal parameter overhead [14]. Addressing the dynamic nature of user profiles over time, Zeng et al. tackled the issue of shifting preferences and catastrophic forgetting associated with global fine-tuning. They introduced the Region-Aware Incremental Editing framework, which clusters historical interactions into distinct knowledge regions. When new behaviors emerge, a confidence-aware routing mechanism triggers localized updates to a specialized LoRA module dedicated solely to the affected region, effectively balancing the assimilation of novel interests with the retention of stable historical behaviors [15].

In domain-specific applications, Guo et al. utilized Quantized LoRA within the TIGMI framework to fine-tune an LLM for cultural tourism. This approach reduced computational overhead by 60 percent while successfully integrating Retrieval-Augmented Generation (RAG) for real-time multimodal interactions [16]. Similarly, Choi et al. applied LoRA to the upper layers of a BERT-based model to enhance a RAG framework for autonomous driving, effectively reducing trainable parameters and improving the retrieval of complex legal texts [17]. Furthermore, Gao et al. demonstrated the scalability of fine-tuned LLMs by developing a multi-dimensional framework to automatically assess the quality and reliability of massive volumes of tourist-generated contents across digital platforms [18].

Conversely, prompt engineering offers an effective alternative that completely avoids altering model weights. Nikolaychuk and Poddubniy applied prompt engineering to the Saiga2 model to automatically extract and classify user sentiments from regional tourism data, successfully achieving high alignment with human expert evaluations [19].

To resolve the complex challenges of intent recognition and factual hallucinations, Huang et al. proposed a multi-layer prompt engineering architecture that decomposes user queries into sequential subtasks. By integrating this step-by-step reasoning strategy with real-time external APIs, their system drastically reduced hallucination rates, proving particularly effective for lightweight models in travel planning applications [20].

Based on the cumulative evidence from previous studies, the goal of this work is to examine how adapter-based PEFT compares with prompt engineering strategies for personalized item recommendation. In particular, the study explores how different adaptation approaches influence ranking accuracy and computational efficiency when utilizing explicit user profiles and structured JSON-formatted catalogs. Guided by prior literature, the following hypotheses are considered:

- H1: PEFT methods may provide improvements in recommendation ranking metrics, such as NDCG and HR, compared to prompt-based adaptation strategies.
- H2: Adapter-based fine-tuning may reduce inference latency and mitigate context-window limitations compared to few-shot prompt engineering.
- H3: LoRA-based adapters may achieve higher ranking accuracy in personalized recommendation tasks than other parameter-efficient techniques such as IA^3.
- H4: Prompt engineering may establish a functional baseline for in-context learning, while PEFT may provide better alignment with explicit user preferences.

3. Methodology

This section details the experimental framework designed to evaluate and compare the efficacy of parameter-efficient fine-tuning (PEFT) against prompt engineering for personalized point-of-interest (POI) recommendation. We outline the data collection process, the synthesis of user profiles for ground-truth annotation, the retrieval-augmented generation (RAG) pipeline, and the specific adaptation strategies applied to the foundation models.

3.1. System Architecture Overview

The proposed recommendation system is evaluated through a RAG framework tailored for structured POI data. The architecture deliberately isolates the retrieval mechanism to ensure a controlled, isolated comparison between two distinct Large Language Model (LLM) adaptation paradigms: Prompt Engineering and PEFT. Three foundation models—Llama 3.1 8B, Qwen 3 4B, and Llama 3.2 3B—are evaluated to assess the scalability, robustness, and architectural generalizability of these adaptation strategies.

3.2. Data Collection and Vector Indexing

To simulate a real-world local recommendation engine, an exhaustive dataset of POIs within the city of Pula, Croatia, was curated via web scraping. The collected data spans multiple domains, including gastronomy, cultural heritage sites, and recreational areas.

The raw data was formalized into a standardized JSON schema. Each POI is represented as a single JSON object containing a categorical type and a nested properties dictionary. To enrich the semantic density of the dataset and improve personalized retrieval, the following key metadata fields were populated:

- Taxonomy Tags: An array of descriptors (e.g., mediterranean, vegetarian) defining the core attributes of the POI.
- Binary and Categorical Constraints: Features such as outdoor_seating, indoor_seating, and price (e.g., low, medium, high) that serve as strict filtering criteria.
- Contextual Narratives: Comprehensive textual descriptions and up to three verified user reviews to provide the embedding model with nuanced sentiment and atmospheric context.

Each JSON object was treated as an independent document chunk. To enable dense retrieval, embeddings were generated using the nomic-embed-text:v1.5 model, selected for its strong performance on heterogeneous text structures. The resulting vectors were indexed using Facebook AI Similarity Search (FAISS) to facilitate efficient similarity computations.

3.3. Synthetic User Profile Generation and Ground Truth Annotation

Evaluating personalized recommendations necessitates a dataset that accurately models diverse user intents and constraints. We synthetically generated 5000 unique user profiles. Each profile represents a distinct permutation of demographic and behavioral traits, mapped to the JSON schema:

- Demographics: gender, age, and nationality to simulate cultural and generational variance.
- Behavioral Archetypes: travel_style (e.g., backpacker, luxury) and persona (e.g., pet_owner) to establish the user's primary motivation.
- Explicit Preferences: likes and dislikes structured as arrays of venue tags to serve as positive and negative alignment vectors.
- Quantitative Constraints: price_preference and min_rating to bound the acceptable POI thresholds.
- Natural Language Query (input): A conversational prompt synthesized from the profile's constraints (e.g., "Any nice outdoor spots where I can bring my pet?").

To establish a high-quality ground truth for training and evaluation, the Qwen 3.6 35B model was utilized to generate initial recommendation responses for all 5000 queries. Because personalized recommendation is inherently subjective, human annotators were employed to blindly review and validate the relevance of the generated recommendations against the explicit constraints of the user profiles.

To ensure the scientific rigor of the dataset, Cohen's Kappa (κ) was calculated to measure inter-rater reliability among the human annotators. Only the subset of responses that achieved absolute consensus (i.e., unanimously marked as correct by all annotators) was retained for the PEFT training corpus. Responses exhibiting annotator disagreement or partial correctness were isolated and repurposed for the benchmark testing dataset, ensuring the models are evaluated on the most challenging, highly subjective edge cases.

3.4. Retrieval and Context Injection Pipeline

During inference, a standardized retrieval pipeline is maintained across all evaluated models to isolate the impact of the LLM adaptation strategy. For a given user input, the query is embedded and searched against the FAISS index.Initially, the vector database retrieves the top $K = 30$ candidate chunks based on vector distance. To maximize the signal-to-noise ratio within the LLM context window, these 30 candidates are subsequently filtered, and only the top 10 most relevant JSON chunks are injected directly into the LLM prompt.

3.5. LLM Adaptation Strategies

The study contrasts a robust Prompt Engineering baseline against two state-of-the-art PEFT methodologies:

- Prompt Engineering (Baseline): The foundation models are augmented with extended, highly structured system instructions. This method relies entirely on in-context learning, leveraging the top 10 retrieved chunks without any underlying weight updates.
- Parameter-Efficient Fine-Tuning (PEFT): The models are fine-tuned on the consensus-driven training set using two distinct adapter mechanisms:

- Low-Rank Adaptation (LoRA): Freezes the pre-trained model weights and injects trainable rank decomposition matrices into the transformer architecture, allowing the model to learn domain-specific ranking logic.
- IA^3 (Infused Adapter by Inhibiting and Amplifying Inner Activations): A highly efficient alternative that scales inner activations using learned vectors, significantly reducing the number of trainable parameters and computational overhead compared to LoRA.

4. Results and discussion

5. Limitations and Future Work

6. Conclusions

This section is not mandatory, but can be added to the manuscript if the discussion is unusually long or complex.

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Abbreviations

The following abbreviations are used in this manuscript:

LLMs	Large Language Models
NLP	Natural Language Processing
PEFT	Parameter-Efficient Fine-Tuning
LoRA	Low-Rank Adaptation
IA^3	Infused Adapter by Inhibiting and Amplifying Inner Activations
ICL	In-Context Learning
JSON	JavaScript Object Notation
SFT	Supervised Fine-Tuning
F1	F1 Score
GPU	Graphics Processing Unit
DDP	Distributed Data Parallel
MLP	Multi-Layer Perceptron
bf16	BFloat16 Precision Format
tf32	TensorFloat-32

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